Reg. No. | | | | | | | | | |

Question Paper Code : 13443

B.E. / B.TECH DEGREE EXAMINATION, NOVEMBER / DECEMBER 2017

Third Semester

B.E. – Computer Science and Engineering

15CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Derive the expressions for Generate and propagate functions of Carry Look ahead Adder.
2. Compare Rounding and Von-Neumann Rounding.
3. Why is the Wait–For–Memory-Function-Completed (WMFC) step needed when reading from or writing to the main memory?
4. Mention the use of control store in micro programmed control.
5. What are pipeline stalls? How they can be overcomed?
6. Differentiate branch taken and branch not taken with an example.
7. Define Double-data-rate SDRAM with an example
8. What is the purpose of Dirty/Modified bit in cache memory?
9. What do you mean by Memory Mapped I/O?
10. What is an Interrupt Service Routine?

PART B – (5 × 16 = 80 Marks)

11. a) i. Define Addressing Mode. Explain the **ADD** instruction by applying various addressing modes found in modern processor. (10 Marks)

11. a) ii. Discuss the non- restoring division algorithm. Simulate the same for 12 / 8. (6 Marks)

OR

11. b) i. Discuss in detail about the design of Full Adder. (10 Marks)

11. b) ii. Perform Carry Save Addition of summands on multiplicand **101101** and multiplier **111111** and state its advantage. (6 Marks)

12. a) i. At some point in the execution of a program, registers R7, R5, and R6 contain the values 3000, 7500, and 2500, respectively. Show the contents of the Multiplexer and the registers RA, RB, RZ, RY, and R6 during steps 3 to 5 as the instruction

Subtract R7, R5, R6 is fetched and executed (10 Marks)

12. a) ii. Discuss in brief the Datapath Organisation of a RISC processor. (6 Marks)

OR

12. b) i. List the sequence of actions needed to execute the following Instructions in terms of data path components

STORE R5,10 (R8)

SUB R3, R4, R5 (10 Marks)

12. b) ii. Write in detail about the interface between processor and memory and the control signals associated with the instruction register. (6 Marks)

13. a) i. Explain how pipelining is done in the CISC style processor with example. (6 Marks)

13. a) ii. **1000 Add R3, R2, #30**

1004 Subtract R5, R4, #35

1008 Add R6, R4, 0X4B

1012 Add R7, R2, R4

Initially, registers R2 and R4 contain 3500 and 50, respectively. These instructions are executed in a computer that has a five-stage pipeline. The first instruction is fetched in clock cycle 1, and the remaining instructions are fetched in successive cycles.

(A) Draw a diagram that represents the flow of the instructions through the pipeline. Describe the operation being performed by each pipeline stage during clock cycles 1 through 8.

(B) With reference to data path describe the contents of registers IR, PC, RA,RB, RY, and RZ in the pipeline during cycles 2 to 8. (10 Marks)

OR

13. b) i. Draw the data hazard diagram and analyze whether operand forwarding mechanism is needed for the following instructions.

- Add R1,R3,R2
  Mul R5,R1,R4
  Add R4,R6,R5
  Add R4,R7,R2
  Add R3.R1.R2
  Sub R6,R1,R5
  Sub R7,R6,R9 (10 Marks)

13. b) ii. List out the ways to reduce the stall in the pipeline due to branch instruction. Discuss static Branch Prediction Technique. (6 Marks)

14. a) i. Explain the use of DMA controllers in a computer system with neat diagram. (6 Marks)

14. a) ii. Design a 64-megabit DRAM chip by organizing it as 8M x 8. Draw the internal organization of the chip and explain its working principle. (10 Marks)

OR

14. b) i. Explain mapping functions in cache memory to determine how memory blocks are placed in cache? (6 Marks)

14. b) ii. Suppose that a computer has a processor with two L1 caches, one for instructions and one for data, and an L2 cache. Let 5τ be the access time for the two L1 caches. The miss penalties are approximately 10τ for transferring a block from L2 to L1, and 30τ for transferring a block from the main memory to L2. For the purpose of this problem, assume that the hit rates are the same for instructions and data and that the hit rates in the

L1 and L2 caches are 0.25 and 0.75, respectively.

(A) What fraction of accesses miss in both the L1 and L2 caches, thus requiring access to the main memory?

(B) What is the average access time as seen by the processor?

(C) Suppose that the L2 cache has an ideal hit rate of 1. By what factor would this reduce the average memory access time as seen by the processor? (10 Marks)

15. a) i. Discuss in detail about PCI and describe how it operates and its features. (8 Marks)

15. a) ii. What are the functions of I/O interface? Explain in detail about Serial interface. (8 Marks)

OR

15. b) i. Explain the bus structure with a neat diagram. (8 Marks)

15. b) ii. With a neat diagram explain the handshake control of data transfer involved in Asynchronous Bus. (8 Marks)